

Q1

Binary Cross Entropy Loss function

Differentiation

Binary cross entropy loss function

$$L = -[y \log(a) + (1-y) \log(1-a)]$$

$$\therefore \frac{dL}{da} = -\frac{d}{da} [y \log(a) + (1-y) \log(1-a)]$$

$$\frac{d}{da} [y \log(a)] = y/a \quad \text{--- (i) [first term]}$$

$$\frac{d}{da} [(1-y) \log(1-a)] = (1-y) \cdot \frac{1 \cdot (-1)}{1-a} = -\frac{1-y}{1-a} \quad \text{--- (ii) [second term]}$$

Substituting in the main equation

$$\begin{aligned} \frac{dL}{da} &= -\left(\frac{y}{a} - \frac{1-y}{1-a}\right) = -\frac{y}{a} + \frac{1-y}{1-a} \\ &= \frac{-y + ay + a - ay}{a(1-a)} \\ &= \frac{a-y}{a(1-a)} \end{aligned}$$

Let

$$a = \sigma(z) = \frac{1}{1+e^{-z}}$$

$$\frac{da}{dz} = a(1-a)$$

Using Chain rule:

$$\frac{dL}{dz} = \frac{dL}{da} \cdot \frac{da}{dz}$$

$$\frac{dL}{dz} = -\log(a) + (1-a)\log(1-a)$$

$$\frac{dL}{dz} = \frac{a-y}{a(1-a)} \cdot a(1-a)$$

$$(1) \quad \frac{dL}{dz} = a-y$$

$$\frac{dL}{dz} = a-y$$

$a = \text{Predicted}$

$y = \text{actual}$

$$\frac{dL}{dz} = \frac{a-y}{a(1-a)}$$

$$\frac{dL}{dz} = \frac{a-y}{a(1-a)}$$

$$\frac{1}{1+e^{-z}} = \sigma(z)$$

$$\sigma(z) = \frac{e^z}{1+e^z}$$